Derivation and Physical Meaning of the Fermi Level in Semiconductors

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1 Origin: Fermi–Dirac Statistics

The Fermi level emerges directly from quantum statistics applied to a system of indistinguishable fermions (electrons) subject to the Pauli exclusion principle. Consider a system of N electrons in thermal equilibrium at temperature T, distributed among a set of available energy states.

1.1 Maximizing entropy under constraints

The equilibrium occupation probability of a state is obtained by maximizing the number of microstates Ω (equivalently, the entropy $S = k_B \ln \Omega$) subject to two constraints:

$$\sum_i n_i = N \quad \text{(fixed total number of particles)}, \tag{1}$$

$$\sum_i n_i E_i = U \quad \text{(fixed total energy)}, \tag{2}$$

where $n_i \in \{0, 1\}$ is the occupation of a single-particle state i with energy E_i (Pauli exclusion: at most one fermion per state, counting spin degeneracy separately).

Using Lagrange multipliers α and β for these two constraints and maximizing the combinatorial entropy of fermionic occupation, one obtains the Fermi–Dirac distribution:

$$f(E) = \frac{1}{e^{\alpha+\beta E} + 1}. \tag{3}$$

Identifying the Lagrange multipliers thermodynamically ($\beta = 1/k_B T$, and $\alpha = -E_F/k_B T$) gives the familiar form:

$$\boxed{f(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{k_B T}\right)}} \tag{4}$$

Here E_F is precisely the Lagrange multiplier associated with the conservation of particle number. This is the key to its physical meaning: E_F is not an "energy level" in the band-structure sense, but a *thermodynamic parameter* conjugate to particle number.

1.2 Thermodynamic identification: the chemical potential

From statistical mechanics, the Lagrange multiplier tied to a conservation constraint on particle number is, by definition, the chemical potential μ:

$$\mu = \left(\frac{\partial U}{\partial N}\right)_{S,V}. \tag{5}$$

Therefore:

$$E_F \equiv \mu(T). \tag{6}$$

At $T = 0\,\mathrm{K}$, μ reduces to the energy of the highest occupied state in the system — the classic "Fermi energy" picture from the free-electron model. At finite T, E_F is more precisely the electron *electrochemical potential*: the energy at which the occupation probability is exactly $1/2$.

2 Physical Meaning

2.1 E_F is not a real energy state

In a semiconductor, E_F frequently lies inside the forbidden gap, where no allowed electronic states exist. This is not a contradiction: $f(E)$ is a probability *function* defined for all E, while the density of states $g(E)$ determines where carriers can actually be found. The carrier concentration is the product:

$$n(E)\,dE = g(E)\,f(E)\,dE. \tag{7}$$

Inside the gap $g(E) = 0$, so no electrons physically sit "at" E_F — yet E_F still meaningfully sets the occupation statistics of the bands above and below it.

2.2 E_F as electrochemical potential / "Fermi level of the system"

Because E_F is a chemical potential, it obeys the same equilibrium principle as in classical thermodynamics: *in any system at thermal and electrical equilibrium, with no net particle flow, the electrochemical potential must be uniform throughout the system.*

This is precisely why, for two materials (or two differently doped regions of the same material) placed in contact and allowed to equilibrate with no applied bias and no net current, a single, spatially flat E_F is established across the whole structure — as discussed for the pn junction. A non-flat E_F would imply $dE_F/dx \neq 0$, which corresponds to a net particle flux (current), contradicting the equilibrium assumption.

2.3 Quantitative link to carrier concentration

For a non-degenerate semiconductor (Boltzmann approximation, valid when E_F is several k_BT away from the band edges), the electron and hole densities are:

$$n = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right), \tag{8}$$

$$p = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right), \tag{9}$$

where N_C, N_V are the effective densities of states of the conduction and valence bands. These relations show explicitly how the position of E_F within the gap controls the carrier population:

- E_F close to $E_C \Rightarrow$ large n (n-type),
- E_F close to $E_V \Rightarrow$ large p (p-type),
- E_F near mid-gap $\Rightarrow n \approx p \approx n_i$ (intrinsic-like).

The product $np = N_C N_V \exp(-E_g/k_BT) = n_i^2$ is independent of E_F — this is the mass-action law, and it is the reason doping shifts E_F rather than changing n_i.

2.4 Degenerate case

At very high doping (N_D or $N_A \gtrsim 10^{19}\,\mathrm{cm}^{-3}$ in silicon), E_F approaches or crosses a band edge, the Boltzmann approximation breaks down, and the full Fermi–Dirac integral must be used. The material then behaves partly like a degenerate electron (or hole) gas, relevant e.g. for heavily doped contact/source regions in power devices.

3 Built-in Potential of the pn Junction

3.1 Setup

Consider an abrupt pn junction formed by joining a uniformly doped n-type region (N_D) and a uniformly doped p-type region (N_A), both non-degenerate. Before contact, each region has its own flat Fermi level:

$$E_C - E_{F,n} = k_B T \ln\!\left(\frac{N_C}{N_D}\right), \tag{10}$$

$$E_{F,p} - E_V = k_B T \ln\!\left(\frac{N_V}{N_A}\right). \tag{11}$$

3.2 Equilibrium condition

After contact, equilibrium requires a single flat E_F across the whole device (Section 2.2). Far from the junction (quasi-neutral bulk regions), the bands retain their original separation from E_F (doping there is unchanged), so the bands as a whole must shift rigidly with respect to each other between the n-side bulk and the p-side bulk. This rigid shift, expressed in volts, *is* the built-in potential φ_0:

$$q\varphi_0 = E_{F,n} - E_{F,p}\Big|_{\text{before contact}}, \tag{12}$$

i.e. the difference between the two original Fermi levels, since both must collapse onto the same final E_F.

3.3 Derivation via carrier concentrations

Using the bulk relations:

$$E_C - E_{F,n} = k_B T \ln\!\left(\frac{N_C}{N_D}\right), \qquad E_{F,p} - E_V = k_B T \ln\!\left(\frac{N_V}{N_A}\right), \tag{13}$$

and noting $E_C - E_V = E_g$:

$$q\varphi_0 = E_g - k_B T \ln\!\left(\frac{N_C}{N_D}\right) - k_B T \ln\!\left(\frac{N_V}{N_A}\right) = k_B T \ln\!\left(\frac{N_D N_A}{N_C N_V}\, e^{E_g/k_B T}\right). \tag{14}$$

Recognizing $n_i^2 = N_C N_V e^{-E_g/k_B T}$ (intrinsic carrier concentration), this collapses to the standard result:

$$\boxed{\varphi_0 = \frac{k_B T}{q} \ln\!\left(\frac{N_D N_A}{n_i^2}\right)} \tag{15}$$

3.4 Equivalent derivation via mass-action law

An alternative, fully equivalent route: in the n-bulk, $n_{n0} \approx N_D$; in the p-bulk, $p_{p0} \approx N_A$. Using $p_{n0} = n_i^2/N_D$ in the n-region and requiring continuity of hole electrochemical potential, one finds the same φ_0 by writing the hole density ratio across the junction:

$$\varphi_0 = \frac{k_B T}{q} \ln\!\left(\frac{p_{p0}}{p_{n0}}\right) = \frac{k_B T}{q} \ln\!\left(\frac{N_A N_D}{n_i^2}\right), \tag{16}$$

identical to the result above. This form makes explicit that φ_0 is exactly the electrostatic potential needed to compensate the diffusion tendency of holes (and, symmetrically, electrons) between the two sides, so that drift and diffusion currents cancel and $dE_F/dx = 0$ everywhere.

3.5 Connection to band bending

φ_0 is not "created" by some external source: it is the electrostatic signature of the flat-E_F condition. The depletion region carries exactly the charge distribution (via Poisson's equation) needed to bend E_C and E_V by $q\varphi_0$ between the two bulks while keeping E_F flat throughout:

$$\frac{d^2\varphi}{dx^2} = -\frac{\rho(x)}{\varepsilon_s}, \qquad \varphi(+\infty) - \varphi(-\infty) = \varphi_0, \tag{17}$$

which, solved with the depletion approximation, also yields the depletion width and the standard 1/2-power voltage dependence used in diode and MOSFET body-junction capacitance formulas — directly relevant for parasitic capacitance estimates in SiC/Si power devices.

4 Summary

- E_F arises mathematically as the Lagrange multiplier enforcing particle-number conservation in Fermi–Dirac statistics.
- Physically, E_F is the electrochemical potential of the electron gas: the energy at which $f(E) = 1/2$.
- E_F need not coincide with an allowed energy state; it is a statistical reference level, not a populated state.
- At equilibrium (no net current, no applied bias), E_F must be spatially uniform across any system — this single principle underlies band bending at junctions and built-in potentials in pn structures, heterojunctions, and metal-semiconductor contacts.
- The position of E_F within the gap directly sets carrier concentrations via the Boltzmann (or, for degenerate doping, Fermi–Dirac) relations.

References

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